

Shyam Agarwal

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EXPERIENCE

- **Software Engineer – Credgenics** July 2024 – Present
Noida, India
 - **Event Streaming:** Own all **Kafka consumers** across the dialer service; primary call-webhook consumer processes **5M+ events/day** with **exactly-once** semantics via application-level idempotency; fan-out to supplementary topics for call-recording sync, webhook forwarding, and borrower-type updates; health monitored via **Grafana** lag dashboards with **KEDA** autoscaling on **AWS EKS**.
 - **Auto Churn:** Designed and owned event-driven lead re-engagement workflow: dialer emits **Kafka lead-cycle-end** event on cycle completion; consumer applies campaign validation logic to identify eligible leads and re-enqueues them for subsequent calling cycles across predictive and voicebot campaigns; independently architected end-to-end; adopted by **85% of campaigns (794/934)** across **142 companies**, covering **2.66M leads** — currently the most adopted optional feature on the platform.
 - **Latency Optimization:** Reduced **p99 latency from 4s to 500ms** on the agent-availability endpoint via Redis caching of per-campaign user slots, composite **PostgreSQL** index tuning, and query batching; validated pre/post release using **Elastic APM**.
 - **DB Reliability & RCA:** Diagnosed and resolved production throughput regression caused by **PostgreSQL row-level lock contention**: identified that batching **campaign_instances** updates with other table writes created hot-row contention under concurrent load; decoupled its execution path from the bulk batch, restoring throughput **same day**; authored RCA shared with DBA and engineering team.
 - **Service Decomposition:** Proposed and led decomposition of monolithic dialer into a dedicated reporting service; authored design doc, received EM sign-off, and configured **KEDA HTTP** autoscaling (**3–30 replicas**) on **AWS EKS** — eliminating **503 errors** on the export consumer and enabling independent scaling of calling and reporting workloads.
 - **Cost Optimization:** Identified accumulation of **7,600+ idle DIDs**; built automated reclamation DAG to detect and return inactive telecom numbers to the allocation pool — avoiding **Rs.22.8L** in new DID procurement on first run, with **800+ DIDs/month** reclaimed ongoing (**Rs.2.4L/month** procurement cost avoided).
 - **API Consolidation:** During vendor-to-in-house dialer migration, consolidated **6 fragmented REST APIs** into **2 unified endpoints**; designed full API contract, ran 30-day parallel production window, deprecated legacy routes — reducing API surface by **67%** and supporting scale from **1M to 5M calls/day**.
 - **Voicebot & Automation:** Integrated backend changes into the in-house LLM voicebot pipeline (**Groq LLM**, self-hosted **TTS**, **STT** vendor APIs); system supports **60 concurrent calls**; authored **5+ Airflow DAGs** (EOD aggregation, DLQ retry, SIP alerting at 15-min cadence); conduct **peer PR reviews** for backend and frontend changes.
- **Software Engineer Intern – Credgenics**  Jan 2024 – Jun 2024
Noida, India
 - **Full-Stack Development:** Built dialer platform UI modules in **ReactJS + MUI**; authored backend **APIs**, **database queries**, and bug fixes for the same calling and campaign management modules.

EDUCATION

- **Indian Institute of Information Technology** Dec 2020 – Jun 2024
Nagpur, India **B.Tech – Computer Science; CGPA: 8.07**

SKILLS

- **Languages:** Python, JavaScript, SQL
- **Backend:** FastAPI, Quartz, asyncio, REST API design
- **Databases & Caching:** PostgreSQL (indexing, query optimization, lock analysis), Redis
- **Messaging & Infra:** Apache Kafka, Redpanda, Airflow, AWS EKS, Kubernetes, KEDA, Docker, Jenkins
- **Observability:** Elastic APM, Grafana
- **Familiar With:** C++, Node.js, ReactJS, Redux, MongoDB

PROJECTS

- **Railway Management System** : Full-stack MERN app with role-based access (guest/user/admin), JWT auth, and ticket booking/cancellation. Stack: **ReactJS, Redux, MongoDB, Node.js, Express.js**.

ACHIEVEMENTS

- **Codeforces:** Rank **2059** in Round 908 Div. 2 (10k+ participants).
- **CodeChef:** Rank **69** in Starters 45 Div. 2; Rank **33** in Starters 41 Div. 3 (both Rated).